

Welcome to the NSF NHERI GSC Natural Hazard Data Challenge!

As part of the *NSF NHERI GSC Mini-Conference*, the Natural Hazard Data Challenge fosters innovative solutions and insights into natural hazard phenomena using datasets from *DesignSafe*. Participants are encouraged to explore freely, apply analytical techniques, and develop machine learning models to address pressing issues in disaster resilience and risk mitigation.

Suggested Datasets and Ideas:

Dataset	Challenge Ideas	Category
Comparing Social Vulnerability and Population Loss in Puerto Rico after Hurricane Maria	1. Use clustering techniques to group census tracts based on social vulnerability index, and other features 2. Identify potential inequities in disaster exposure and preparedness if it is possible 3. Use findings to recommend targeted interventions for vulnerable tracts	Social/ Hurricane
Predicting Real-time Population Behavior during Hurricanes Synthesizing Data from Transportation Systems and social media	1. Analyze traffic behavior across different time periods (e.g., Early Morning, Evening) to understand how evacuation strategies can be tailored to specific times of day. 2. Study how cumulative population data under mandatory evacuation orders correlates with traffic congestion and evacuation speed 3. Use insights from the dataset to propose actionable recommendations for improving disaster preparedness and evacuation strategies.	Hurricane/ Tropical Storm
Dataset - Experimental studies on rehabilitation techniques in masonry walls	1. Compare the experimental results in the dataset to predictions from theoretical models 2. Create a decision-making framework to guide practitioners in selecting the best rehabilitation technique based on the dataset 3. Analyze the performance of different rehabilitation methods (e.g., steel bands vs. FRP bands). 4. Identify which techniques are most effective in improving the structural integrity of masonry walls.	Earthquake/ Structure
A Unified Multi-Event Windstorm Performance Testbed for Single-Family Residential Buildings	1. Compare the damage patterns caused by hurricanes and tornadoes 2. Develop a machine learning model to predict component-level damage ratios based on building attributes and wind speeds. 3. Highlight clusters of high damage and potential contributing factors. 4. Explore which building attributes (e.g., roof type, construction material) are most strongly correlated with higher damage ratios.	Wind

Criteria:

1. **Poster Submission:** Teams must submit a poster showcasing their work to the NHERI NSF GSC Mini-Conference.
2. **Dataset:** Only datasets provided by the DesignSafe Data Depot may be used for the challenge.
3. **Mandatory Check-In:** At least one member from each team must attend the designated data challenge check-in session.
4. **Sharing Work:** While not mandatory, participants are encouraged to share their code or analysis in the DesignSafe folder specifically created for the mini conference.
5. **Bonus Consideration:** Projects that utilize datasets to address social problems will receive bonus recognition.

Participant Benefits:

- **Winner Recognition:** Certificate of achievement.
- **Presentation Opportunity:** Winners will present via a recorded 15-minute video published on YouTube.
- **Mentorship Insights:** Feedback from the mentors will be provided during the check-in sessions

Timeline:

- ✓ Data Challenge Check-in#1: February 28
- ✓ Data Challenge Check-in#2: March 28
- ✓ Data Challenge Check-in#3: April 25